Self-Service Coffee Kiosk — UML Component Diagram (Layered Architecture)

«layer» Presentation Layer

Customer

«use»

ITouchInput
Touch-screen events

«component»
Touch Screen UI
menu screens · size picker · checkout

IOrder
API call (in-process)

«layer» Business Logic Layer

«component»
Payment Service
card-only checkout

IPayment
API call: processPayment()

«component»
Order Manager
builds order · computes total

IReceipt
API call: printReceipt()

«component»
Receipt Service
formats order details

ICardAuthorization
HTTPS / TLS (PCI-DSS)

IMenuData
Database query (SQL)

IPrinterDriver
Hardware interface: USB / ESC-POS

«layer» Data & Hardware Access Layer

«external»
Card Payment Gateway
bank / card processor

«component»
Menu & Pricing Repository
3 coffees × 2 sizes, prices

«device»
Receipt Printer
thermal printer hardware

Legend

Provided interface (ball) · Required interface (socket) · Port · «use» dependency · Blue text = protocol / technology of the connector